# Intro to Data Science - Lab 6

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## Week 6 - Using ggplot to Build Complex Data Displays

```
# Enter your name here: Tim Soron
```

**Please include nice comments.**

**Instructions:**

Run the necessary code on your own instance of R-Studio.

**Attribution statement: (choose only one and delete the rest)**

```
# 1. I did this lab assignment by myself, with help from the book and the professor.
```

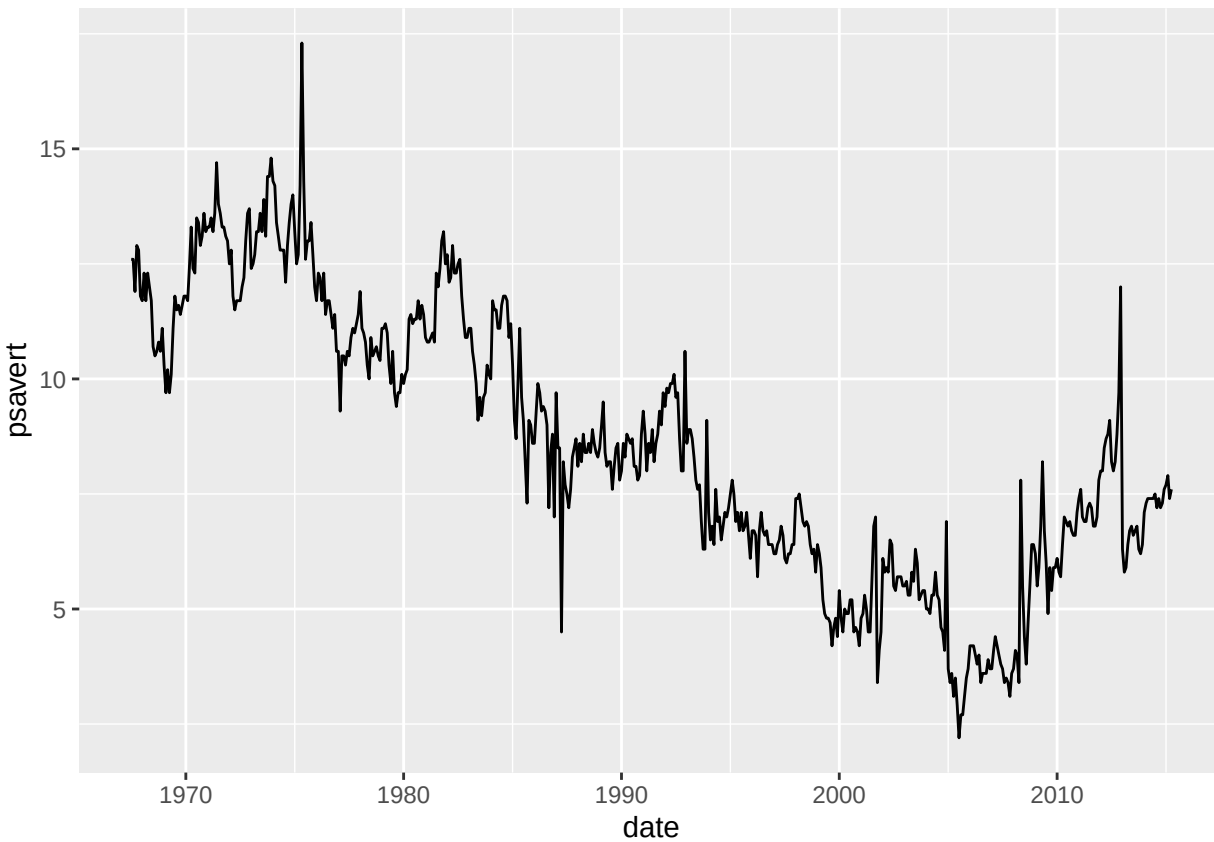
Creating graphical displays of data is an essential skill for all data scientists because so much of what data scientists do involves communicating with other people about data. The **ggplot2** package developed by data scientist Hadley Wickham (https://en.wikipedia.org/wiki/Hadley_Wickham) provides excellent power and flexibility for graphically displaying data. Whole books have been written about **ggplot2** (e.g.: https://www.springer.com/gp/book/9780387981413 ), so we will only be able to scratch the surface, but we will master the basic grammar that you need in order to use this
package. For this lab we will also use a dataset that comes delivered in R, called ** economics **. It contains 574 snapshots of U.S. economic conditions between 1967 and 2014.
Here are two lines of starter code:

```
#install.packages('ggplot2')
library("ggplot2")
```

```
MyPlot <- ggplot(economics, aes(x=date))
myPlot <- myPlot + geom_line(aes(y=psavert))
```

1. Run these two lines of code below. What happens? How do you actually invoke the plot (i.e., how do you get it to draw in the plot window)?

```
ggplot(economics, aes(x=date, y=psavert))+
  geom_line()
```




2. Run `help("economics")` to find out the meaning of the **psavert** variable.

```
#help(economics)
economics_df <- economics
```

3. Examine the plot to estimate when the personal savings rate reached its maximum value. Also examine the plot to estimate when the personal savings rate reached its minimum value.

```
# max was at 1975, min was at 2006
```

4. Use **which.max( )** and **which.min( )** to verify your guesses from problem 3.

```
economics[which.max(economics$psavert), "date"]
```

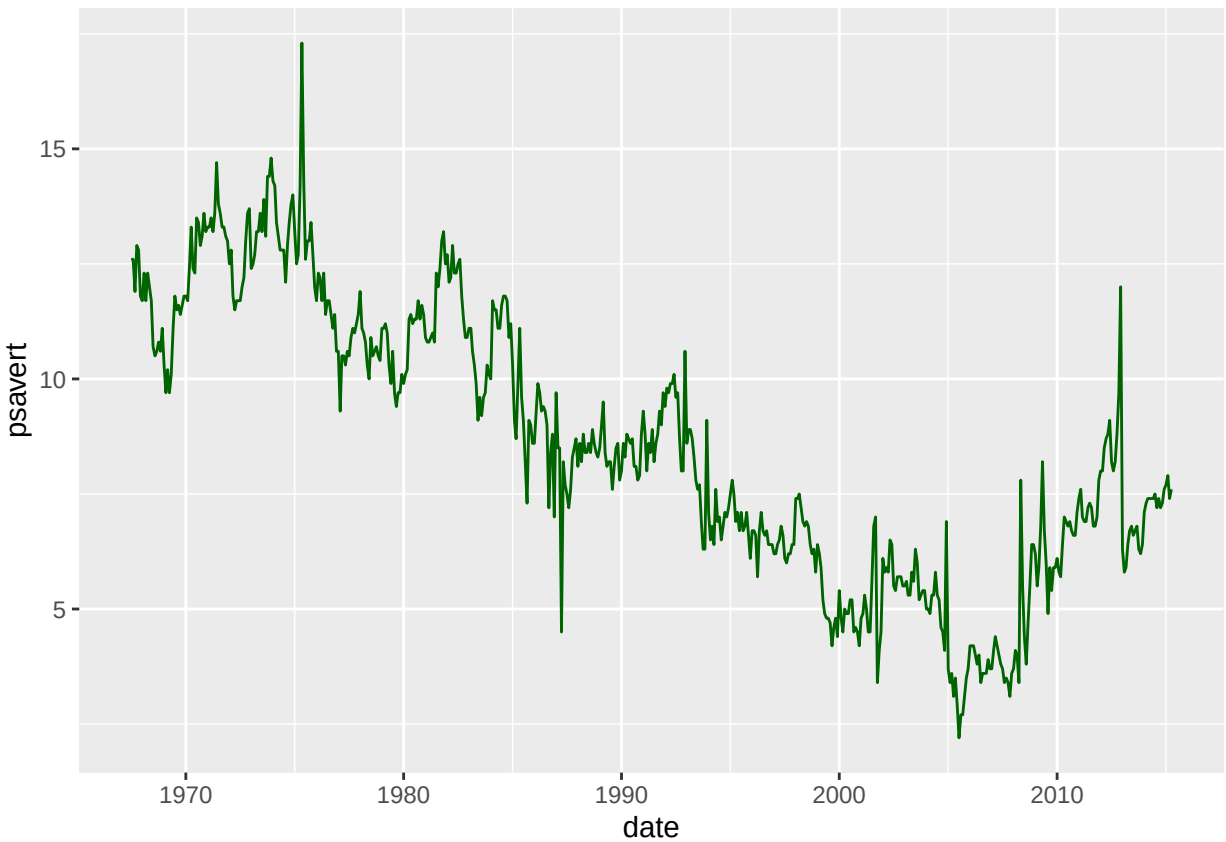
```
## [1] "1975-05-01"
```

```
economics[which.min(economics$psavert), "date"]
```

```
## [1] "2005-07-01"
```

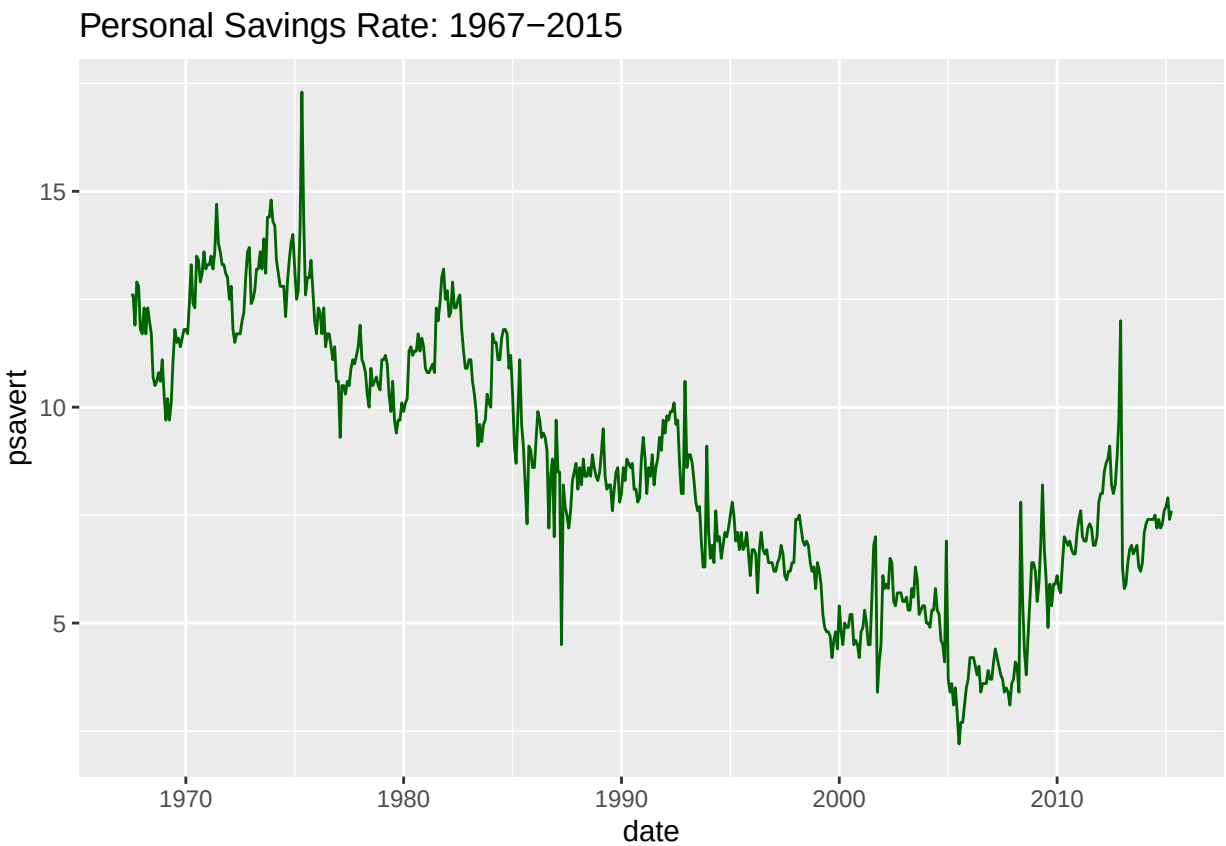
5. Change the color of the plot line to green. **Hint:** Changing a line to a constant color happens in the specification of the **geometry**.

```
ggplot(economics, aes(x=date, y=psavert))+
  geom_line(color='darkgreen')
```



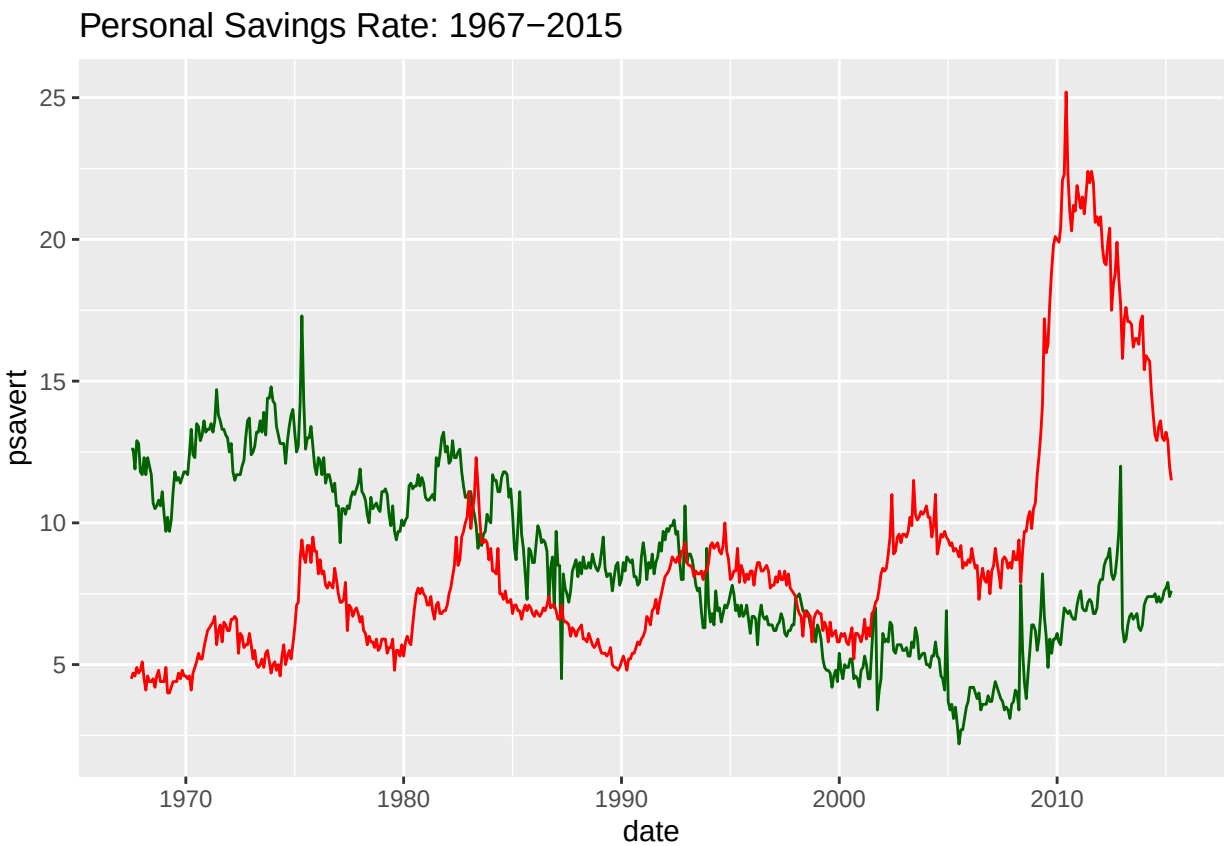

6. Add a title to the plot with the **ggtitle("Put title here")** sub-command. The title **"Personal Savings Rate: 1967-2014"** would be a good choice.

```
ggplot(economics, aes(x=date, y=psavert))+
  geom_line(color='darkgreen')+
  ggtitle("Personal Savings Rate: 1967-2015")
```



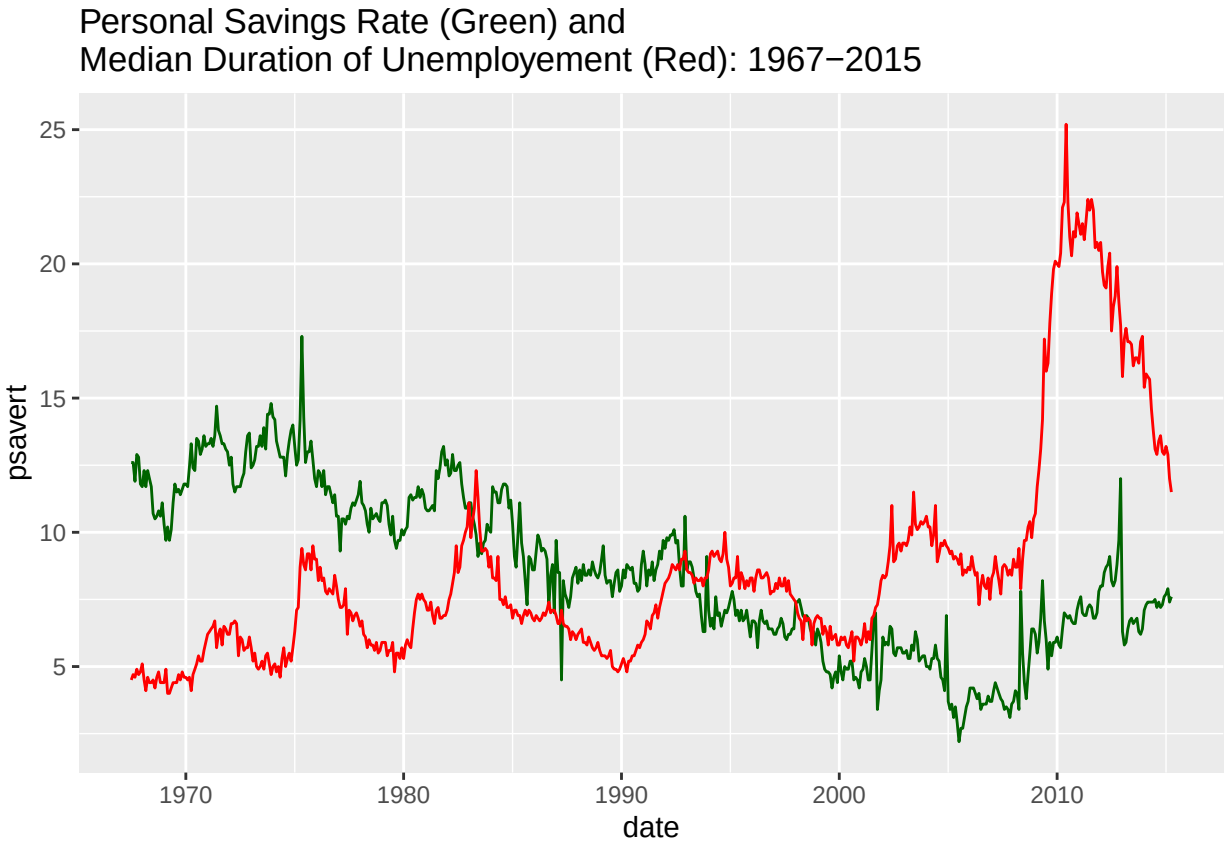

7. Add another data series to your plot to show the variable **uempmed** as a red line.

```
ggplot(economics, aes(x=date, y=psavert))+
  geom_line(color='darkgreen', aes(y=psavert))+
  geom_line(color='red', aes(y=uempmed))+
  ggtitle("Personal Savings Rate: 1967-2015")
```



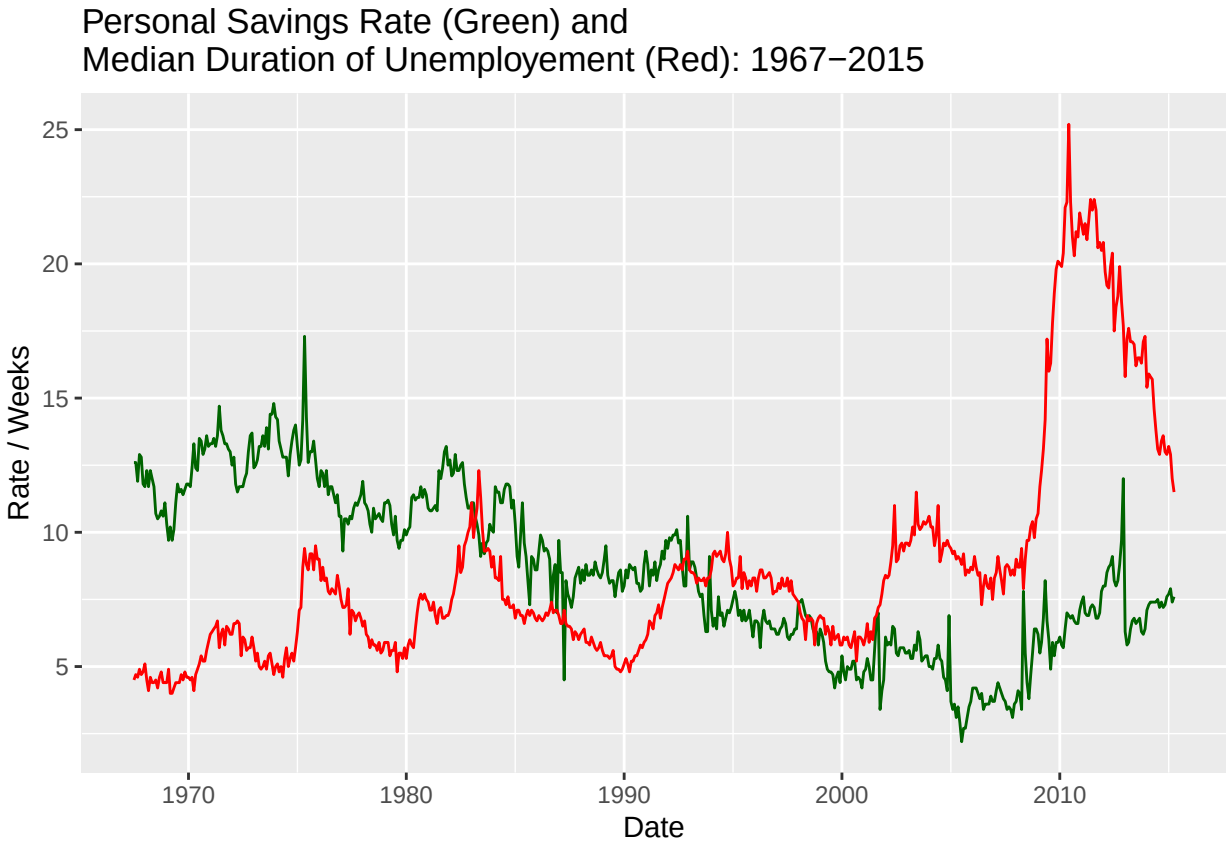

8. Change the title of the plot to mention both variables.

```
ggplot(economics, aes(x=date, y=psavert))+
  geom_line(color='darkgreen', aes(y=psavert))+
  geom_line(color='red', aes(y=uempmed))+
  ggtitle("Personal Savings Rate (Green) and \nMedian Duration of Unemployement (Red): 1967-2015")
```



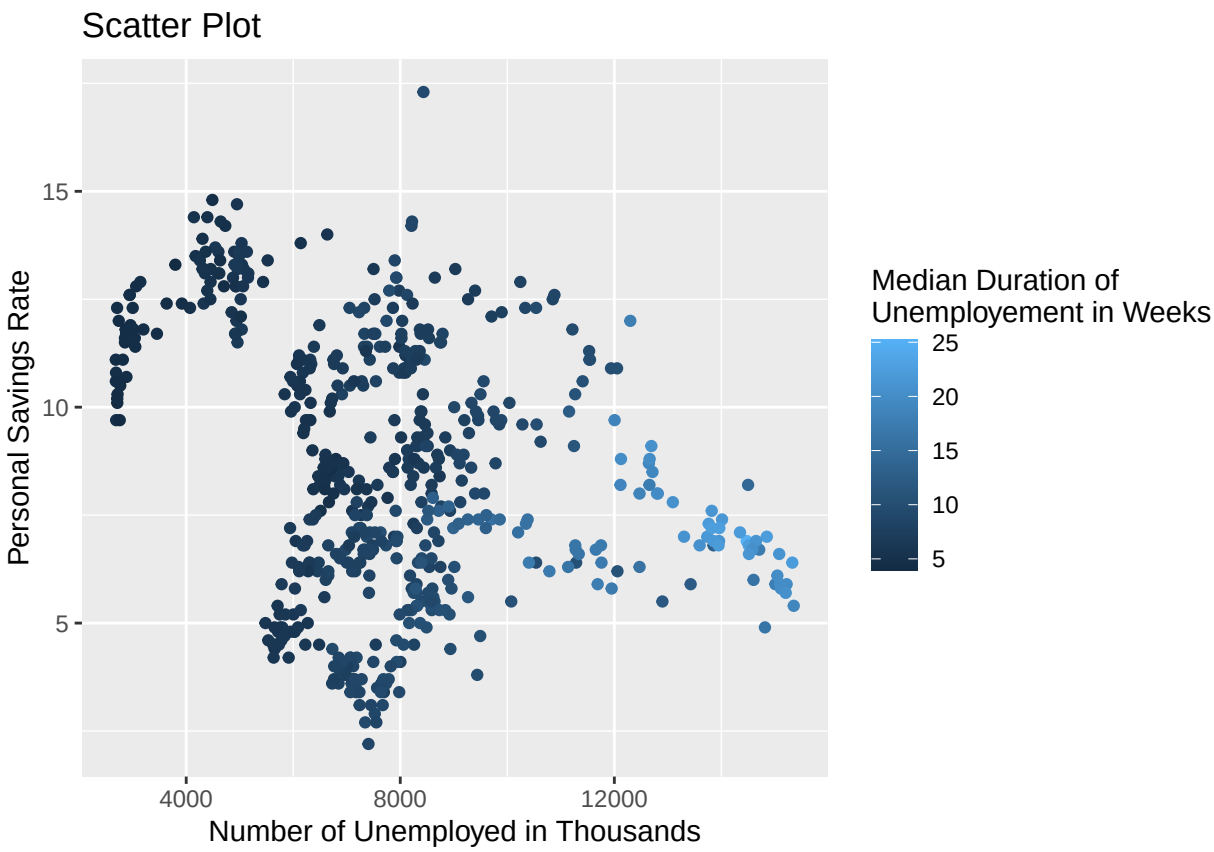

9. You can modify the axis labels in a ggplot with **ylab( )** and **xlab( )** subcommands. Change the axis labeling as needed to account for plotting both **psavert** and **uempmed** in the same window.

```
ggplot(economics, aes(x=date, y=psavert))+
  geom_line(color='darkgreen', aes(y=psavert))+
  geom_line(color='red', aes(y=uempmed))+
  ggtitle("Personal Savings Rate (Green) and \nMedian Duration of Unemployement (Red): 1967-2015")+
  xlab("Date")+
  ylab("Rate / Weeks")
```




10. Create one last plot, creating a scatter plot, having the **unemploy** on the x-axis, **psavert** on the yaxis. Color each point based on the **uempmed**.

```
ggplot(economics, aes(x=unemploy, y=psavert))+
  geom_point(aes(color=uempmed))+
  ggtitle("Scatter Plot")+
  xlab("Number of Unemployed in Thousands")+
  ylab("Personal Savings Rate")+
  labs(color= "Median Duration of \nUnemployement in Weeks")
```




11. Interpret (using comments in R) what you see in this last graph.

```
# The two variables have somewhat negative of a relationship which makes sense as more unemployment in
```